

Since the **location** column contains commas (e.g., "New York, USA"), we should use **OpenCSV SerDe** to correctly handle quoted fields.

Create an External Table to Read Data Directly with Create Partitioned Table (Partitioned by Date)

```
CREATE EXTERNAL TABLE iot_sensor_data (  
    sensor_id STRING,  
    timestamp STRING,  
    temperature FLOAT,  
    humidity FLOAT,  
    pressure FLOAT,  
    location STRING, -- Properly handles "New York,USA"  
    recent_temperatures ARRAY<FLOAT>,  
    metadata MAP<STRING, STRING>  
)  
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'  
WITH SERDEPROPERTIES (  
    "separatorChar" = ",",  
    "quoteChar" = "\"" )  
STORED AS TEXTFILE  
LOCATION '/mnt/data/';
```

The company wants to **reduce storage costs** and **improve query speed** using compression.

```
SET hive.exec.compress.output=true;  
SET mapred.output.compression.codec=org.apache.hadoop.io.compress.SnappyCodec;  
SET hive.exec.orc.default.compress=ZLIB;
```

```
CREATE TABLE iot_sensor_compressed (  
    sensor_id STRING,  
    timestamp STRING,  
    temperature FLOAT,  
    humidity FLOAT,  
    pressure FLOAT  
)  
STORED AS ORC;
```

1. SELECT location, AVG(temperature) AS avg_temp, AVG(pressure) AS avg_pressure

```
FROM iot_sensor_data
GROUP BY location
HAVING avg_temp > 35 AND avg_pressure < 1000;
```

```
2. SELECT sensor_id, COUNT(*) AS duplicate_count
FROM iot_sensor_data
GROUP BY sensor_id, timestamp
HAVING COUNT(*) > 1;
```

```
3.
SELECT sensor_id, COUNT(*) AS data_points
FROM iot_sensor_data
GROUP BY sensor_id
ORDER BY data_points DESC
LIMIT 10;
```

```
4.
SELECT sensor_id, recent_temperatures[SIZE(recent_temperatures)-1] AS
last_temp_reading
FROM iot_sensor_complex;
```

```
5.
SELECT sensor_id, metadata["manufacturer"] AS manufacturer
FROM iot_sensor_complex
WHERE metadata["manufacturer"] = 'XYZ Inc.';
```

```
6.
SELECT sensor_id, device_info.installation_date
FROM iot_sensor_complex
WHERE device_info.installation_date < '2024-01-01';
```

```
7.
SELECT sensor_id, recent_temperatures
FROM iot_sensor_complex
WHERE recent_temperatures[SIZE(recent_temperatures)-1] -
recent_temperatures[SIZE(recent_temperatures)-2] > 5;
```

```
8.
SELECT sensor_id
FROM iot_sensor_complex
WHERE metadata["manufacturer"] IS NULL;
```

```
9.
SELECT location, MAX(pressure) - MIN(pressure) AS pressure_variation
FROM iot_sensor_complex
GROUP BY location
```

HAVING pressure_variation > 10;

10

```
SELECT sensor_id,  
       MAX(recent_temperatures[0], recent_temperatures[1], recent_temperatures[2],  
recent_temperatures[3], recent_temperatures[4]) -  
       MIN(recent_temperatures[0], recent_temperatures[1], recent_temperatures[2],  
recent_temperatures[3], recent_temperatures[4])  
       AS temp_fluctuation  
FROM   iot_sensor_complex  
HAVING temp_fluctuation > 5;
```

11

```
SELECT sensor_id, metadata["manufacturer"] AS manufacturer, metadata["model"] AS  
model, temperature, humidity  
FROM   iot_sensor_complex  
WHERE  metadata["manufacturer"] IN ("IoT Corp", "SensorTech")  
      AND metadata["model"] IN ("X100", "Y200");
```

12.

```
SELECT sensor_id, sensor_status.last_service_date AS last_active_date  
FROM   iot_sensor_complex  
WHERE  sensor_status.active = FALSE  
      AND sensor_status.last_service_date < DATE_SUB(CURRENT_DATE, 30);
```